Computers & Geosciences 171 (2023) 105270

Contents lists available at ScienceDirect

Computers and Geosciences

journal homepage: www.elsevier.com/locate/cageo

SigMT: An open-source Python package for magnetotelluric data processing

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ARTICLE INFO

Keywords:
SigMT
Python package
Data processing
Magnetotellurics

ABSTRACT

The magnetotelluric (MT) data processing is often a time-consuming job due to the manual inspection of the time series and removal of noisy segments. Use of different data selection tools combined with the robust estimation of the impedances has enabled the automation of MT data processing to a large extent. In this paper, we introduce an open-source Python package, named 'SigMT' for the automated MT data processing. This Python package offers Python scripts to estimate MT impedances and tipper data from the raw time series data. The overview of the SigMT package explaining different steps involved in the processing is discussed in this paper. Different data selection tools such as Mahalanobis distance based-, coherency threshold based-, polarization direction based-selection tools are integrated in the package. The incorporation of different data selection tools with robust estimation technique delivers best estimates of the impedances. SigMT is applied to the MT data from Himalaya with different selection tools and found to be yielding satisfactory results.

1. Introduction

Urbanization is a biggest challenge nowadays for the application of magnetotelluric (MT) method as it introduces cultural noises and results in the limited availability of locations for the data acquisition. This scenario demands advancements in the signal processing techniques and better understanding of the electromagnetic signals to obtain the desired quality of the MT data. Different branches of MT data processing have evolved over the time. The robust estimation of the MT impedance (Egbert and Booker, 1986; Chave et al., 1987; Egbert, 1997; Ritter et al., 1998; Smirnov, 2003) overruled the use of least square estimation, which is used to get biased even with very few bad data points. The remote reference method which involves the simultaneous recording of magnetic components from different locations helps to remove the effect of uncorrelated noises in the recordings (Goubau et al., 1978; Gamble et al., 1979). Since, the strong outliers bias the robust estimation, different data selection criteria such as selective stacking, coherency threshold (Swift, 1967; Jones and Jödicke, 1984), polarization direction (Weckmann et al., 2005) are commonly used prior to the robust estimation. Artificial neural networks is shown to be effective in removing noisy segments in the time series (Manoj and Nagarajan, 2003; Imamura and Schultz, 2020) and that helps to reduce the manual interventions in time series editing. The Mahalanobis distance (MD) based pre-selection tool (Platz and Weckmann, 2019) offers automated data selection in a very effective way. These different data selection tools are sufficient in most of the cases except in unavoidable instances such as acquisition near highly populated cities, near electric power production sites, etc., where complete dataset is usually contaminated.

In this paper, we introduce a Python package named 'SigMT' for the MT data processing. It is an open-source Python package which helps to process the raw MT time series data to estimate MT impedances and tipper data in an automated manner. At present, this package supports '.ats' file format for the processing. But support for more file formats will be added in the upcoming updates. There are different academically available codes for the estimation of the MT transfer function such as EMTF (Egbert and Booker, 1986; Egbert, 1997) and Bounded Influence Remote Reference Processing (BIRRP) (Chave and Thomson, 2004) which are based on the Fortran programming language. But the MTPy software package (Krieger and Peacock, 2014) enables the use of BIRRP package in Python environment. MTPy offers a unified software environment, where a person without much programming skills can also experience different operations in MT such as processing, analysis, visualizations, and modelling. Smaï and Wawrzyniak (2020) presented the Razorback Python package for the robust estimation of the MT impedances. Razorback Python package delivers both M-estimator and bounded influence methods and multiple remote reference. Similarly, the

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https://doi.org/10.1016/j.cageo.2022.105270
Received 10 June 2022; Received in revised form 9 November 2022; Accepted 9 November 2022
Available online 16 November 2022
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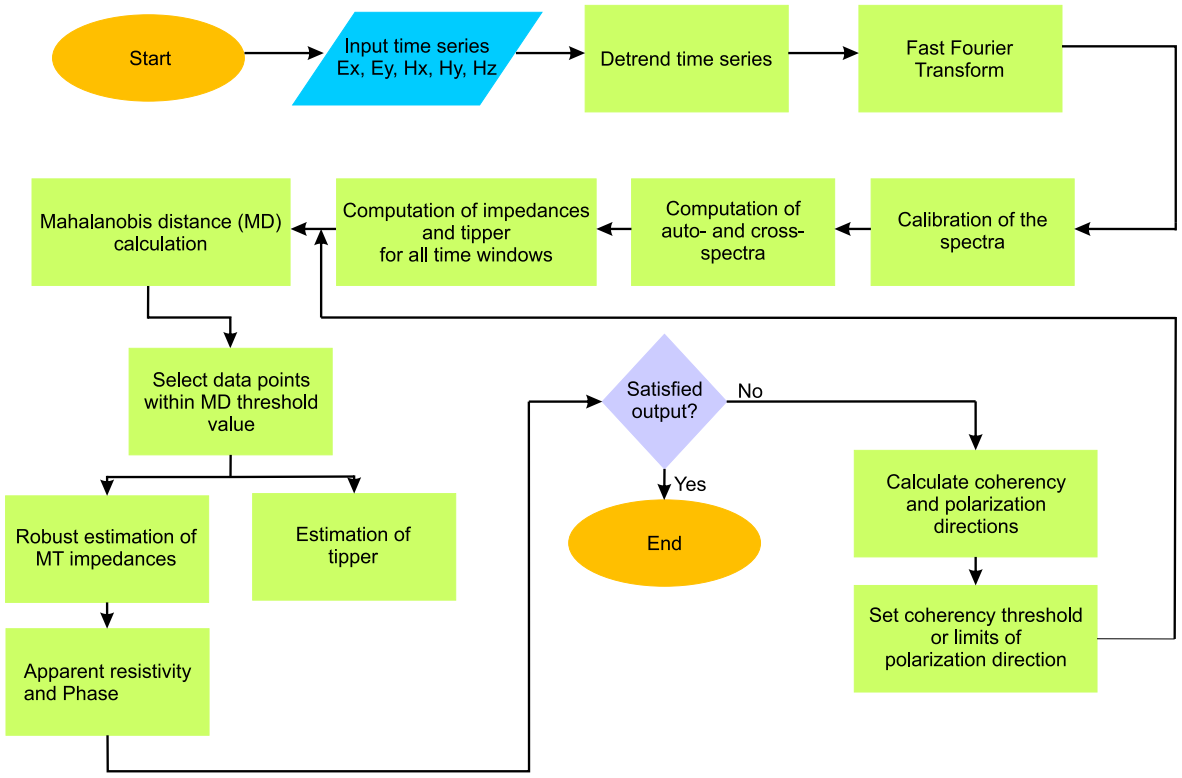


Fig. 1. Flow chart showing the overview of SigMT package.

Resistics[1] Python package also enables the robust estimation of the MT impedances. Another option is the windows based free software, ElectroMagnetic Processor (EMP)[2] by Arseny Shlykov. The SigMT stands apart from these alternates as it gives additional options for different data selection tools. Also, it is organized in such a way that one can easily add more features to the existing package and can perform different experiments.

2. Magnetotelluric method

Magnetotellurics (MT) is a passive electromagnetic method which uses natural electromagnetic (EM) signals as the source field (Vozoff, 1991; Simpson and Bahr, 2005; Chave and Jones, 2012). The natural EM fields are mainly generated by the interaction of solar wind and cosmic rays with the ionosphere and because of the worldwide thunderstorm activities. The source of these naturally occurring electromagnetic fields are far apart from the earth surface and so, the interaction of these fields with the earth surface is considered as plane waves. After interacting with the earth surface, the input magnetic field (H) diffuses through the earth and produces orthogonal (E) secondary electric field in the Earth following the electromagnetic induction phenomenon. These orthogonal magnetic and electric fields are combined through the transfer function - impedance (Z). The impedance is the ratio between electric (E) and magnetic (H) fields. They are generally expressed as a complex impedance tensor as follows (in frequency domain):

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{bmatrix} \begin{bmatrix} H_x \\ H_y \end{bmatrix} \tag{1}$$

The impedance is associated with both amplitude and phase as it is a complex value. Amplitude provides the apparent resistivity (ρ_a) and phase (φ) gives the difference between the instantaneous phases of H and E for each frequency. The apparent resistivity (ρ_a) and phase (φ) are related to the impedance as:

$$\rho_a = \frac{1}{\omega \mu_0} |Z|^2 \tag{2}$$

$$\varphi = \tan^{-1}\left(\frac{Im\,(Z)}{Re\,(Z)}\right) \tag{3}$$

$\omega = angular\ frequency, \mu_0 = permeability\ of\ free\ space$

Similarly, the vertical (H_z) and horizontal (H_x and H_y) components of the magnetic field are related through the transfer function called tipper (T). It is expressed as (in frequency domain):

$$H_z = [\,T_x \quad T_y\,] \begin{bmatrix} H_x \\ H_y \end{bmatrix} \tag{4}$$

Tipper can be used to define the induction vectors which helps to understand the lateral conductivity variations. Inclusion of tipper data in modelling studies enables the improved horizontal resolution for the conductive bodies.

The magnetotelluric method is widely used for tectonics studies, hydrocarbon explorations, mineral and geothermal investigations (Jones et al., 2001; La Terra and Menezes, 2012; Patro 2017; Samrock et al., 2018; Egbert et al., 2022).

During the MT data acquisition, the electric field (E) and magnetic field (H) components are continuously recorded as time series at the observation locations. During the recording, some segments of the time series may be affected by noises. The noises are mainly the electromagnetic signals from different artificial sources (e.g., human interference, powerlines, electrical equipment, etc.) that are added with the desired natural electromagnetic signal. The removal of noise and the estimation of MT impedances from the raw time series data acquired from the observation location is generally termed as 'MT data processing'. Three orthogonal magnetic field components (H_x : *north* − *south*, H_y : *east* − *west*, H_z : *vertical*) and two orthogonal electric field

[1] https://resistics.io/.
[2] https://www.csrmt.info/emp-main-eng.

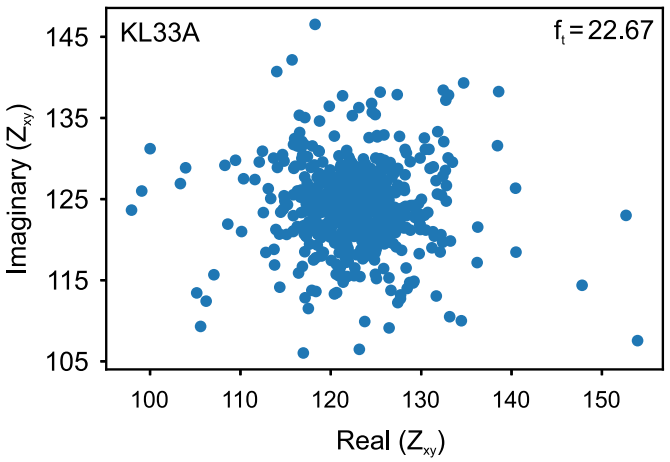


Fig. 2. Argand diagram showing the impedances of xy component for all time windows.

components ($E_x : north - south, E_y : east - west$) are measured from each MT sites. The magnetic field is measured using the induction coil magnetometers (for broadband MT) and fluxgate magnetometers (for long period MT). The porous pot electrodes are used to make a dipole and measure the electric field. In areas of high contact resistance, metal plates are also used to measure the electric field.

3. Overview of SigMT

The package is created as a project in the Spyder[3] IDE with the support of Anaconda[4] Python distribution. Spyder is a free and open-source scientific environment for Python. A flow chart showing the different steps involved in this Python package is shown in Fig. 1. The input files are the raw time series data for five channels such as E_x, E_y, H_x, H_y and H_z. SigMT presently supports '.ats' file format from the ADU 07e (Metronix Geophysics) data logger. The '.ats' file is a binary file which contains the raw time series data with header information. Initially, the time series data is read from the binary files. Then the raw time series is corrected for the linear trend. The detrended time series is converted to the frequency domain using the Fast Fourier Transform (FFT). The window length (n) for FFT is selected as per the criteria proposed by Borah et al. (2015). Otherwise, there is an option for the user to define FFT length. In the case of broadband MT data, the magnetic field and electric field data outputs are generated in milli volts (mV). Hence, the magnetic field data need to be converted to nano Tesla (nT) using the calibration values given for the induction coil magnetometers. The Fourier transform will generate a series of values for n/2 frequencies. But we limit our observations for target frequencies considering the diffusive nature of the MT signal. The MT impedance are slowly varying functions in the frequency domain, hence the values of auto- and cross-spectral values for each target frequencies are averaged using a parzen window (Jenkins and Watts, 1968). Parzen window has an advantage that it gives less weightage to the values for frequencies away from the target frequency. Parzen window is defined as,

$$W(f)\begin{cases} 1, if\ |f_t - f| = 0 \\ \left(\frac{\sin(u)}{u}\right)^4, if\ 0 < |f_t - f| < f_r \\ 0, if\ |f_t - f| \geq f_r \end{cases} \quad (5)$$

where $u = \pi|f_t - f|/f_r$

[3] https://www.spyder-ide.org/.
[4] https://www.anaconda.com/.

f_t is the target frequency, f_r is the radius of the parzen window. The parzen window radius is selected according to Borah et al. (2015). Else, user can define the parzen window radius. Once the auto- and cross-spectral values are computed, the MT impedance values can be calculated for all the time windows using the following equations.

$$Z_{xx} = \frac{\langle H_y H_y^* \rangle \langle E_x H_x^* \rangle - \langle H_y H_x^* \rangle \langle E_x H_y^* \rangle}{\langle H_x H_x^* \rangle \langle H_y H_y^* \rangle - \langle H_x H_y^* \rangle \langle H_y H_x^* \rangle} \quad (6)$$

$$Z_{xy} = \frac{\langle H_x H_x^* \rangle \langle E_x H_y^* \rangle - \langle H_x H_y^* \rangle \langle E_x H_x^* \rangle}{\langle H_x H_x^* \rangle \langle H_y H_y^* \rangle - \langle H_x H_y^* \rangle \langle H_y H_x^* \rangle} \quad (7)$$

$$Z_{yx} = \frac{\langle H_y H_y^* \rangle \langle E_y H_x^* \rangle - \langle H_y H_x^* \rangle \langle E_y H_y^* \rangle}{\langle H_x H_x^* \rangle \langle H_y H_y^* \rangle - \langle H_x H_y^* \rangle \langle H_y H_x^* \rangle} \quad (8)$$

$$Z_{yx} = \frac{\langle H_x H_x^* \rangle \langle E_y H_y^* \rangle - \langle H_x H_y^* \rangle \langle E_y H_x^* \rangle}{\langle H_x H_x^* \rangle \langle H_y H_y^* \rangle - \langle H_x H_y^* \rangle \langle H_y H_x^* \rangle} \quad (9)$$

Asterisk indicates complex conjugate.

Similarly, the tipper values can be computed using the following equations.

$$T_x = \frac{\langle H_z H_x^* \rangle \langle H_y H_y^* \rangle - \langle H_z H_y^* \rangle \langle H_y H_x^* \rangle}{\langle H_x H_x^* \rangle \langle H_y H_y^* \rangle - \langle H_x H_y^* \rangle \langle H_y H_x^* \rangle} \quad (10)$$

$$T_y = \frac{\langle H_z H_y^* \rangle \langle H_x H_x^* \rangle - \langle H_z H_x^* \rangle \langle H_x H_y^* \rangle}{\langle H_x H_x^* \rangle \langle H_y H_y^* \rangle - \langle H_x H_y^* \rangle \langle H_y H_x^* \rangle} \quad (11)$$

Once the impedances are calculated for each target frequencies for all time windows, the Mahalanobis distance is calculated for each time window.

The Mahalanobis distance (MD) (Mahalanobis, 1936) based data selection is very useful in MT data processing to separate noisy time windows in an automated way (Platz and Weckmann, 2019). For a particular target frequency, there exist only one impedance at an observation point. If we plot an argand diagram of impedance values for all time windows for a particular target frequency (Fig. 2), we can observe a cluster, which represent the impedance at the observation point. The values away from the cluster are outliers and can be rejected. The rejection of the outliers away from the cluster can be done using the MD. The MD of the ith observation is calculated using the equation (Platz and Weckmann, 2019),

$$MD_i = \sqrt{(x_i - \mu) C_x^{-1} (x_i - \mu)^T} \quad (12)$$

x_i is the multivariate measurement, expressed as real and imaginary components of impedances at a frequency in rows. For example, Re(Z_{xx}), Im(Z_{xx}), Re(Z_{xy}), Im(Z_{xy}). μ is the data centre of the distribution, and C_x is the covariance matrix.

The robust estimation of the data centre and the covariance matrix is obtained using the Minimum Covariance Determinant (MCD) algorithm. Once the data centre and covariance matrix are calculated, the MD for each observation can be calculated using equation (12). The computations to obtain Mahalanobis distance are done using the Scikit-learn (Pedregosa et al., 2011) Python library. The data points fall within the MD threshold limit (For example: MD < 1.5) are selected for the robust estimation of impedance values and all others time windows are rejected. Similarly, the tipper values are averaged using the events within the MD threshold limit.

The final estimation of the impedance value is carried out using the robust estimation method. The robust estimation is a branch of statistics, which is designed to work on distributions having moderate quantity of outliers. The least square (LS) method works well on data having normal distribution of errors, but in MT, the robust estimation is preferred, as it is insensitive even to the non-Gaussian errors. There are different robust statistical estimators are available, but the M-estimator (maximum likelihood) estimator is used in the MT data processing. It is an itera-

tively reweighted LS algorithm that does not allow few data points to control the estimate using the down weighting procedure (Egbert and Booker, 1986). We use huber M-estimator in this package for the robust estimation. To begin the iteration, an initial value of impedance is required. The initial guess of impedances is obtained using the Jackknife estimator (Efron, 1982; Chave and Thomson, 1989). It is given by,

$$Z_{jackknife} = L \times Z^{mean} - \frac{L-1}{L} \sum_{i=1}^{L} Z_{-i} \tag{13}$$

Where, L is the number of estimates of Z, Z^{mean} is the mean of all impedances, Z_{-i} is the mean rejecting the ith observation of Z.

The formulation of robust estimation explained hereafter is based on Ritter et al. (1998) and Manoj (2003). Once the initial guess of impedance is calculated, the residuals can be computed using the equation,

$$r_i^l = \left| E_i^l - Z_{ii} H_i^l - Z_{ij} H_j^l \right| \tag{14}$$

(i = x or y)

A mean absolute deviation (MAD) scale estimate is used as an initial estimate for the robust weighting process. It is given by,

$$d_M = 1.483 \times median\left(\left| r^l - median\left(r^l\right)\right|\right) \tag{15}$$

$$k_M = 1.5 \times d_M \tag{16}$$

Then the huber weights w^l for each time windows are calculated using the equation,

$$w^l = \begin{cases} 1, for & r^l \le k_M \\ \frac{k_M}{r^l}, for & r^l > k_M \end{cases} \tag{17}$$

Using the above huber weights, the auto- and cross-spectra can be averaged, for example,

$$\langle \overline{E_x H_y^*} \rangle = \frac{1}{\sum_{l=1}^{L} w^l} \sum_{l=1}^{L} w^l \langle E_x H_y^* \rangle^l \tag{18}$$

After each iteration, a new MAD scale estimator is calculated using equation,

$$d_H^2 = \frac{1}{L_c^2} \sum_{l=1}^{L} w^l \left(r^l\right)^2 \tag{19}$$

After sufficient iterations, the final impedance values can be calculated by substituting auto- and cross-spectra in equations (6)–(9).

The statistical variance for impedances is calculated using the equation (Manoj, 2003),

$$\Delta Z_{xy}^2 = k \frac{F(k, 2N-4, \delta = 0.05)\left[1 - \gamma_{E_x E_x^p}^2\right] E_x E_x^*}{(2N-4)\left[1 - \gamma_{H_x H_y}^2\right] H_y H_y^*} \tag{20}$$

F is Fisher distribution (Bendat and Piersol, 2010). k is 4 and N is the number of Fourier coefficients used and 2N-4 is the degree of freedom, $\delta = 0.05$ corresponding to 95% confidence limit, γ is predicted coherence (Equation (24)).

The apparent resistivity and phase can be calculated from the impedance values using equations (2) and (3). The error bars are calculated using the following equations,

$$\Delta \rho_{ij} = \frac{2\mu}{\omega} \left|Z_{ij}\right|^2 \Delta Z_{ij} \tag{21}$$

$$\Delta \varphi_{ij} = \frac{\Delta Z_{ij}}{\left|Z_{ij}\right|} \tag{22}$$

(i,j = x or y)

ΔZ is the variance.

In this Python package, the option for the remote reference method is also included. The remote reference method (Goubau et al., 1978; Gamble et al., 1979) is a commonly used data processing technique in MT. It is useful in the cases where local magnetic field values are affected by noise. For this purpose, during field program, a site which is known to be unaffected by any noise is continuously recorded throughout the field season. Shalivahan and Bhattacharya (2002) discussed how far a remote station can be installed to improve the quality of MT data. In this method, the magnetic field data collected from the remote site (Rx and Ry) are used as complex conjugates during the impedance estimation. For example,

$$Z_{xy} = \frac{\langle \overline{H_x R_x^*} \rangle \langle \overline{E_x R_y^*} \rangle - \langle \overline{H_x R_y^*} \rangle \langle \overline{E_x R_x^*} \rangle}{\langle \overline{H_x R_x^*} \rangle \langle \overline{H_y R_y^*} \rangle - \langle \overline{H_x R_y^*} \rangle \langle \overline{H_y R_x^*} \rangle} \tag{23}$$

But this method cannot yield better results if the electric field data is contaminated.

After the estimation of impedances, the apparent resistivity and phase for xy- and yx-components are plotted with the error bars. If the output is not satisfactory, the further analysis and rejection of noisy segments can be done using the coherency values and polarization directions.

3.1. Coherency threshold based selection

Coherency is a parameter which represent the linear relationship between two data sets. The coherency function is generally used in MT to find the linear relationship between measured (E) and predicted (E^p) output electric field data, termed as predicted coherency. The equation to calculate the predicted coherency is given by (modified after Swift, 1967):

$$\gamma_{E_i^p E_i} = \frac{\langle E_i^p E_i^* \rangle}{\sqrt{\left(\langle E_i^p E_i^{p*} \rangle \langle E_i E_i^* \rangle\right)}} \tag{24}$$

(i = x or y)

where,

$$E_i^p = Z_{ix} H_x + Z_{iy} H_y \tag{25}$$

The predicted coherency value is usually used to understand the certainty of the estimated impedances. Its value lies in a range of (0,1). If the estimated impedance is in line with the observed E and H fields, then the coherency takes the value of close to 1.0. If any of the channels (E or H) is affected by the noise (incoherent noise), the value of coherency will be less than the 1.0. The coherency can also be near to 1.0, in the cases of the presence of noise in both the E and H channels or if any process other than electromagnetic induction is involved. In general, low coherency value depicts the presence of more noises in the data. The coherency function can be used for the data selection prior to the robust estimation (Jones and Jödicke, 1984). The impedance values that show low coherency than a threshold value can be rejected from further processing to obtain best approximation of impedances during robust estimation.

3.2. Polarization direction based selection

Polarization direction based selection is another efficient tool to understand the noise in the MT data. The polarization directions of electric field and magnetic field in MT are given by (Fowler et al., 1967; Weckmann et al., 2005):

$$\alpha_E = \tan^{-1}\left(\frac{2 \bullet Re\left(\left[E_x E_y^*\right]\right)}{\left[E_x E_x^*\right] - \left[E_y E_y^*\right]}\right) \tag{26}$$

```
config['FFT_Length'] = 0
# Give '0' to automatically select an FFT length according to
# Borah et al. (2015)
# Otherwise try 256, 512, 1024, 2048, 4096, 8192, 16384
config['parzen_radius'] = 0
# Give '0' to select default parzen radius
config['MD_threshold_impedance'] = 1.5
# Mahalanobis distance threshold value for impedance calculations
config['MD_threshold_tipper'] = 1.0
# Mahalanobis distance threshold value for tipper calculations
```

Fig. 3. Script to set different processing parameters.

```
ts, procinfo['fs'], procinfo['sensor_no'], timeline, \
    procinfo['ChoppStat'], loc = mtproc.ts(proc_path)
```

Fig. 4. Script to read time series data. The input is the file path and outputs are time series data and few information about the measurement.

```
ftlist,bandavg = mtproc.bandavg(ts,procinfo,config)
```

Fig. 5. Script to perform the band averaging over target frequencies. Inputs are time series (ts), measurement information (procinfo) and configuration information (config). Outputs are the target frequency list (ftlist) and auto and cross spectra values over target frequencies (bandavg).

$$\alpha_B = \tan^{-1}\left(\frac{2 \bullet Re\left(\left[B_x B_y^*\right]\right)}{\left[B_x B_x^*\right] - \left[B_y B_y^*\right]}\right) \tag{27}$$

The natural magnetic field is expected to have varying polarization directions with respect to time as it is generated from different sources such as interaction of solar wind with ionosphere, ionospheric currents, and worldwide thunderstorm activities (Weckmann et al., 2005). However, the electric field can be polarized in a particular direction based on the subsurface conductivity distribution. But, if noise is interfered with the signal, the measured magnetic and electric field might be polarized in a particular direction. Weckmann et al. (2005) utilized this property to categorize noise affected time windows and natural signal time windows. Best impedance values can be estimated after rejecting the polarized time windows.

The Mahalanobis distance will be computed again using the time windows satisfying the coherency threshold and/or polarization direction condition and then the robust estimation is carried out. In noisy environments, there can be two clusters in the data (Platz and Weckmann, 2019) that will mislead the Mahalanobis distance calculation. But this issue can be solved using the polarization direction based data selection. One of the clusters, which represents the noisy time windows may show a particular polarization. So, rejecting time windows with that particular polarization direction will take out the noisy cluster.

4. Some example codes

4.1. Setting of different parameters

SigMT provides an option to set different processing parameters such as FFT length, Parzen window radius, Mahalanobis distance threshold for impedance and tipper calculations (Fig. 3). The Python dictionary 'config' is used to save the information. The default value for FFT length and Parzen radius need to be kept '0' to select default values according to Borah et al. (2015). Else, provide the desired value for the corresponding dictionary key. Similarly, users can edit the Mahalanobis distance threshold values for impedance and tipper calculations.

4.2. Reading of the time series

Fig. 4 shows the script to read time series. The function 'ts' is written in 'mtproc' module to read the time series data and header information from the files. The input 'proc_path' is the path to the folder where raw time series data files are stored. The E_x, E_y, H_x, H_y and H_z components will be read from the corresponding files and will be saved in the Python dictionary 'ts'. Some information such as sampling frequency (fs), sensor numbers, time and date of samples, etc. are saved in the 'procinfo' Python dictionary. The location and elevation details of the site will be saved in the 'loc' Python dictionary.

4.3. Band averaging over target frequencies

Fig. 5 presents the script to perform band averaging over the target frequencies. The averaging of E_x, E_y, H_x, H_y, H_z, auto- and cross-spectra over target frequencies as well as calculation of impedance values for all time windows for all target frequencies are carried out in the 'bandavg' function in the 'mtproc' module. The Python dictionaries ts, procinfo and config are the inputs. The 'ts' dictionary contains the raw time series data. The 'procinfo' contains some information such as FFT length, sensor numbers, etc. The parzen window radius information which is required during this process is carried in the 'config' Python dictionary. The list of target frequencies (ftlist), auto and cross spectra values and impedance values for all time windows (bandavg) are returned from the function.

4.4. Coherency calculation

The coherency values are calculated for each time window for all target frequencies using the script shown in Fig. 6. The 'data_sel' module is used for this operation. Coherency of E_x and E_y components can be calculated using the 'cohEx' and 'cohEy' functions respectively. The coherency values are saved as arrays in 'AllcohEx' and 'AllcohEy'. These values will be used for the coherency threshold based data selection.

4.5. Polarization direction calculation

The polarization directions are calculated using the function

```
123   AllcohEx = data_sel.cohEx(bandavg)
124   AllcohEy = data_sel.cohEy(bandavg)
```

Fig. 6. Script to calculate coherency values. The input is 'bandavg'Python dictionary and outputs are coherency values of Ex (AllcohEx) and Ey (AllcohEy) components.

```
126   alpha_degH,alpha_degE = data_sel.pdvalues(bandavg)
```

Fig. 7. Script to calculate polarization directions. The input is the 'bandavg' Python dictionary containing auto and cross spectra values and output is the polarization directions for magnetic (alpha_degH) and electric (alpha_degE) field components.

```
129 ctflag = 0 # Give '1' to perform coherency threshold based selection
130 if ctflag == 1:
131     CohThre = [0.9,0.9,0.9,0.9,0.9,0.9,0.9,0.9,0.9,0.9,0.9,0.9,0.9,0.9,0.9,0.9]
132     for i in range(np.shape(AllcohEx)[0]):
133         for j in range(np.shape(AllcohEx)[1]):
134             if AllcohEx[i,j] < CohThre[i]:
135                 cohMatrixEx[i,j] = 0
136             else:
137                 cohMatrixEx[i,j] = 1
138             if AllcohEy[i,j] < CohThre[i]:
139                 cohMatrixEy[i,j] = 0
140             else:
141                 cohMatrixEy[i,j] = 1
```

Fig. 8. Script to perform coherency threshold based data selection. 'ctflag' need to be kept '1' to perform the task. Threshold values can be given in the 'CohThre'. 'cohMatrixEx'and 'cohMatrixEy' are the outputs.

```
144 pdflag = 0 # Give '1' to perform polarization direction based selection
145 if pdflag == 1:
146     pdlim = [-10,10]
147     alpha = alpha_degE # Use either alpha_degE or alpha_degH
148     for i in range(np.shape(pdmat)[0]):
149         for j in range(np.shape(pdmat)[1]):
150             if alpha[i,j] > pdlim[0] and alpha[i,j] < pdlim[1]:
151                 pdmat[i,j] = 0
152             else:
153                 pdmat[i,j] = 1
```

Fig. 9. Script to perform the polarization direction based selection. 'pdflag' need to be kept '1' to perform the task. The polarization direction limits can be given in 'pdlim'. The output is 'pdmat'.

```
155 pdflag = 0 # Give '1' to perform polarization direction based selection
156 if pdflag == 1:
157     timewindow_limits = [0,5]
158     for i in range(np.shape(pdmat)[0]):
159         for j in range(np.shape(pdmat)[1]):
160             if j > timewindow_limits[0] and j < timewindow_limits[1]:
161                 pdmat[i,j] = 0
162             else:
163                 pdmat[i,j] = 1
```

Fig. 10. Script to perform the polarization direction based selection. 'pdflag' need to be kept '1' to perform the task. The time window limits can be given in 'timewindow_limits'. The output is 'pdmat'.

'pdvalues' (Fig. 7). The 'pdvalues' function is included in the 'data_sel' module. It requires the 'bandavg' dictionary containing auto- and cross-spectral values. The function returns the polarization directions for magnetic field (alpha_degH) and electric field (alpha_degE) for all time windows for every target frequency.

```
182    [Tx, Ty] = tipper.tipper(bandavg)
183    [TxVar, TyVar] = tipper.tipperVar(bandavg)
```

Fig. 11. Script for tipper estimation and variance calculation. Input is the auto and cross spectra values (bandavg). Outputs are tipper values (Tx and Ty) and tipper variance (TxVar and TyVar).

```
189    Z_huber = mtproc.perform_robust(ftlist,bandavg)
190    Zvar['xx'],Zvar['xy'],cohEx = var.ZExvar(Z_huber,bandavg)
191    Zvar['yx'],Zvar['yy'],cohEy = var.ZEyvar(Z_huber,bandavg)
```

Fig. 12. Script for the impedance estimation and variance calculation. The inputs for impedance estimation are target frequency list (ftlist) and auto and cross spectra (bandavg). Output is the huber estimates of impedance (Z_huber). The inputs for variance are impedance values (Z_huber) and auto and cross spectra (bandavg). Outputs are variances (Zvar) and predicted coherency for Ex and Ey components (cohEx and cohEy).

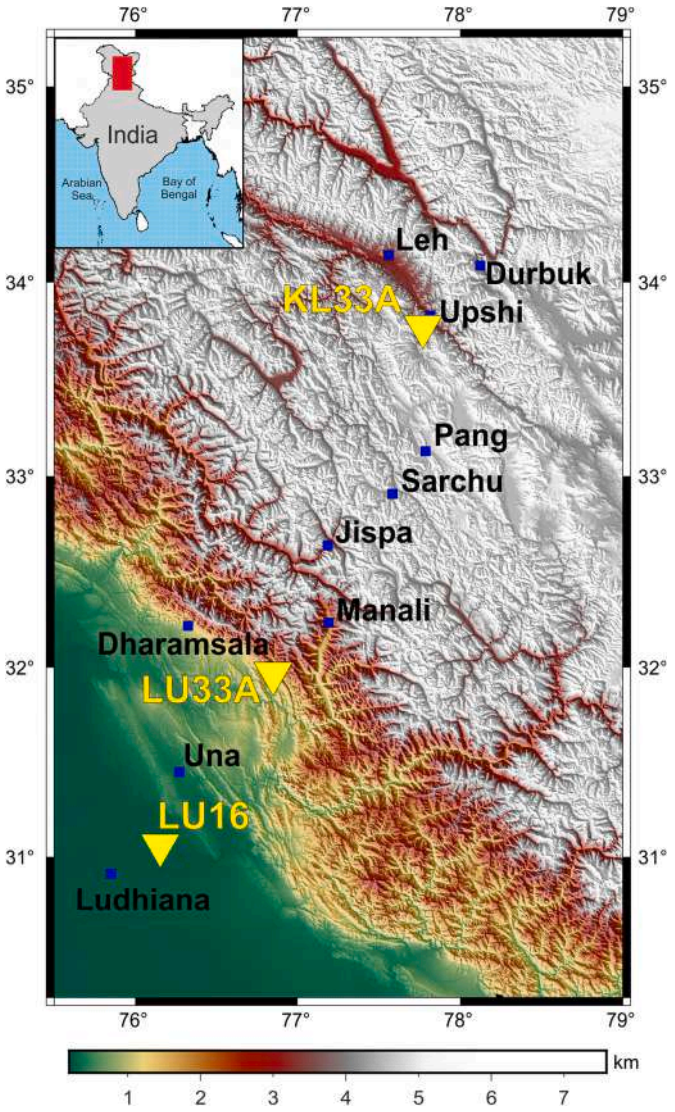


Fig. 13. Topographic map showing the locations of the MT sites. Left top inset: Key map of the study area.

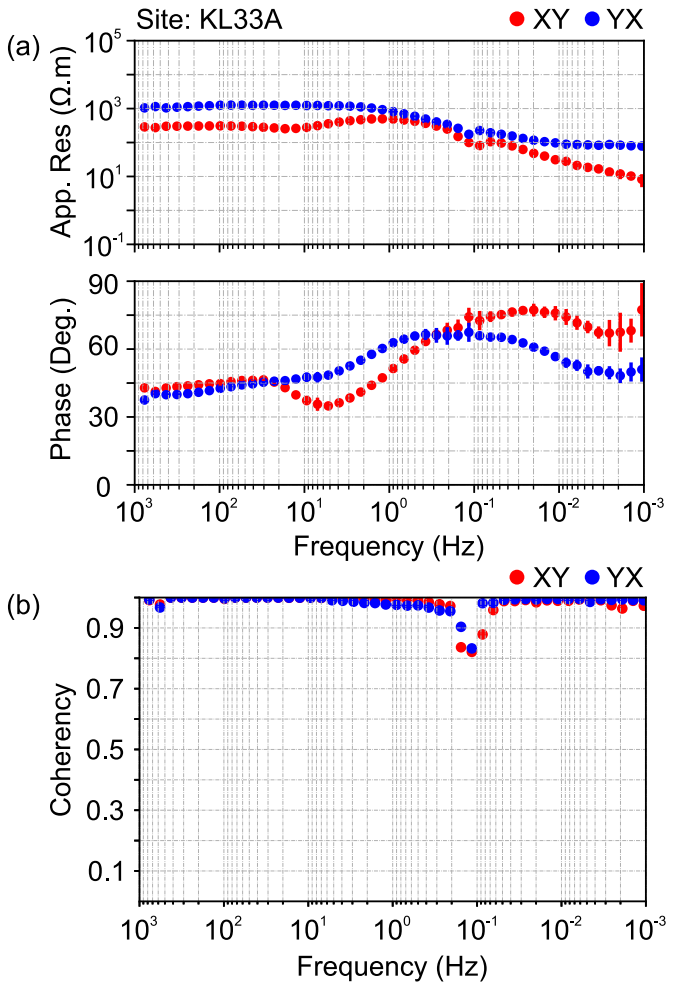


Fig. 14. (a) Apparent resistivity and phase for the site KL33A. (b) Predicted coherency for the site KL33A.

4.6. Coherency threshold based data selection

The coherency threshold based data selection is performed using the code segment shown in Fig. 8. The 'ctflag' should be kept '1' to perform the data selection. Give threshold values in the variable 'CohThre' corresponding to each target frequencies. Number of threshold values should be equal to the number of target frequencies. In the example (Fig. 8), we have given threshold value of 0.9 for all target frequencies. Then the code will make two matrices for Ex (cohMatrixEx) and Ey (cohMatrixEy) components with values of '0' and '1'. The time windows with coherency values less than the threshold value will be given a value of '0' and others are given value of '1'. The time windows with value '1' will be selected for the further processing and windows with values '0' will be rejected.

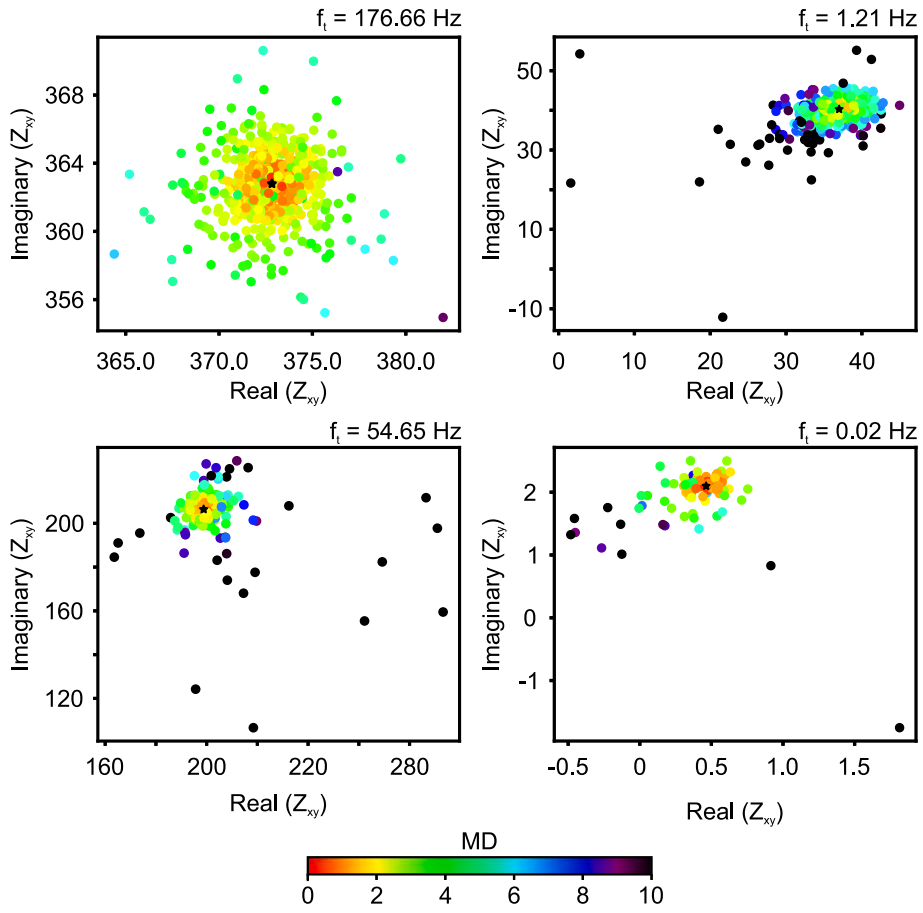


Fig. 15. Mahalanobis distance (MD) for the target frequencies (f_t) 176.66 Hz, 54.65 Hz, 1.21 Hz and 0.02 Hz. The data center is shown as dark solid star.

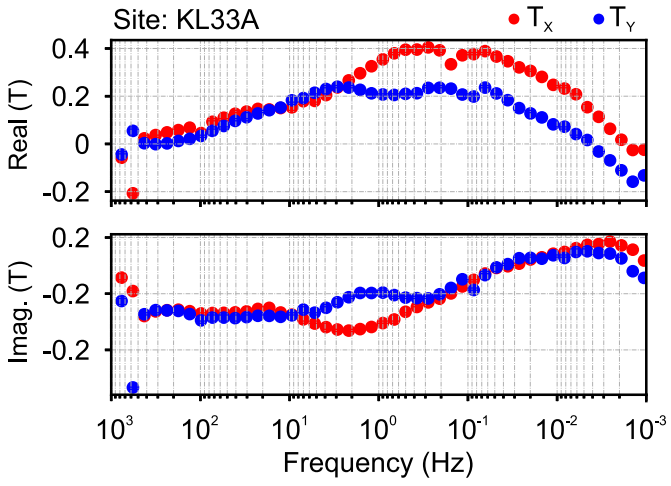


Fig. 16. Tipper data.

4.7. Polarization direction based data selection

The polarization direction based data selection can be done in two ways. In the first method, we can reject time windows falls in a range of polarization direction (Fig. 9). For example, we can reject time windows having polarization direction in a range of (−10, 10). We can either use polarization directions for magnetic field (alpha_degH) or electric field (alpha_degE). The code will create a matrix 'pdmat' with values '0' and '1'. The time windows having polarization values in the limit will be given a value of '0', that means rejection. Similarly, other time windows will be given a value of '1' which means selection.

In the second method (Fig. 10), we can reject a portion of time windows. Give values of time window limits in the variable 'time-window_limits'. For example, if we need to reject first time window to the 50th one, give limit as [0,49].

4.8. Tipper estimation and calculation of variance

Fig. 11 shows the scripts to perform tipper estimation and variance calculation. The tipper components T_x and T_y are estimated using the 'tipper' function in the 'tipper' module. The tipper values for each target frequency are estimated using the time windows selected using the Mahalanobis distance based selection. The 'tipperVar' function returns the variances for T_x and T_y components.

4.9. Robust estimation of the impedances and calculation of variance

Fig. 12 shows the scripts to perform robust estimation of impedances and variance calculation. The 'perform_robust' function in 'mtproc' module is used to perform robust estimation of the impedance values. It calls other function to perform the task and returns the huber estimates ('Z_huber') of the impedance value for every target frequency. The 'ZExvar' and 'ZEyvar' functions in the 'var' module that computes the variances for the impedance values and predicted coherency values (cohEx and cohEy) for each target frequency. Variances for Z_{xx} and Z_{xy} components are calculated using the 'ZExvar' function and variances for Z_{yx} and Z_{yy} components are calculated using the 'ZEyvar' function.

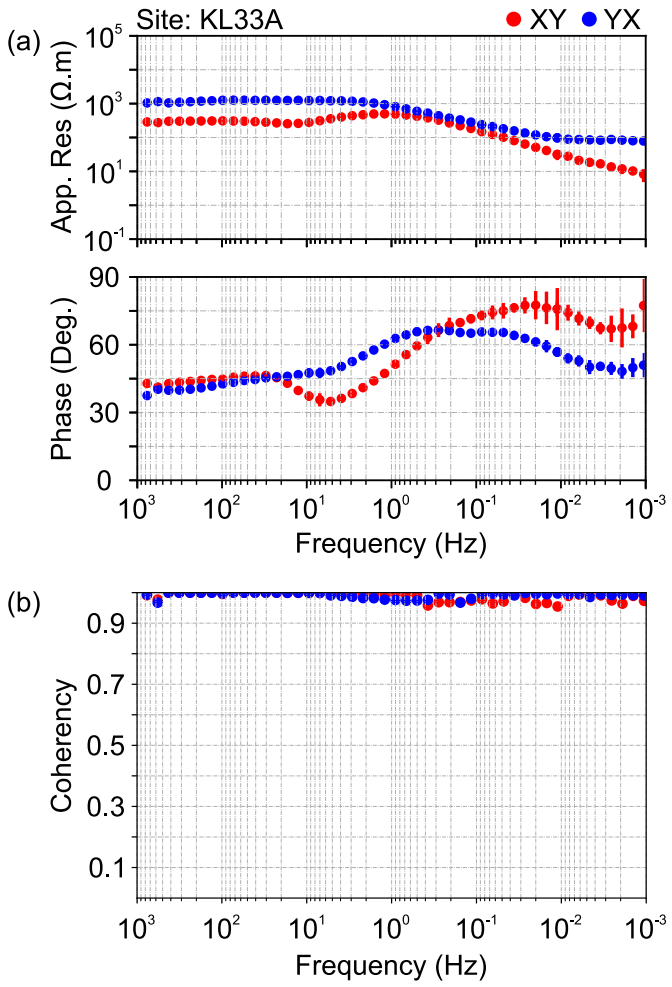


Fig. 17. (a) Apparent resistivity and phase for the site KL33A after applying the remote reference method. (b) Predicted coherency for the site KL33A after applying remote reference method.

5. Application to real data from Himalaya

The Himalayan Mountain range is formed as a result of the northward drift of Indian plate in Tethys Sea and collision with the Eurasian plate during ~50 Mya (Molnar and Tapponnier, 1975; Copley et al., 2010). Hence, the Himalaya is an interest of study in terms of the continent – continent collision tectonics, mountain building process and the earthquakes. MT method is a very useful geophysical method to study the lithospheric structures. MT has a demonstrated history in imaging the crustal weak zones and geo-fluids (Ogawa et al., 2001; Unsworth et al., 2005; Becken et al., 2011; Patro and Sarma, 2016). To understand the subsurface structures and the associated tectonic process, we acquired broadband magnetotelluric (BBMT) data and long period magnetotelluric (LMT) data during the 2018–2019 field campaign in the NW Himalaya. Here, we present the processing of three BBMT sites (KL33A, LU33A, LU16) to show the efficiency of the SigMT package. The locations of the sites are shown in Fig. 13.

We acquired broadband MT data using different sampling frequencies such as 65536 Hz, 4096 Hz, 1024 Hz, 512 Hz, 128 Hz and 32 Hz to get a wide band of data from 1000 Hz to 0.001 Hz. The lower frequencies are obtained by decimating the 32 Hz time series. In Fig. 14a, we present smooth curves of apparent resistivity and phase of XY and YX components from the site KL33A. Fig. 14b represents the predicted coherency for both XY and YX components. Coherency is near to 1.0 in most of the values for both components except for the frequencies near to 0.1 Hz. Drop in the coherency values in those frequencies are due to the dead band and presence of noise in data. Small error bars in the phase near to 0.1 Hz are due to the drop in the coherency values. As per equation (20), the variance increases when the coherency values are less. Error bars are also seen in the phase values of low frequencies (<0.003 Hz), which is due to the low degree of freedom in lower frequencies. Fig. 15 demonstrates the impedances values (xy component) for all time windows and its Mahalanobis distance (MD) for the target frequencies 176.66 Hz, 1.21 Hz, 54.65 Hz and 0.02 Hz for KL33A. The events with MDs less than 1.0 are only selected for the robust estimation of the data. In Fig. 15, it may see that, the extreme outliers as well as data points away from data centre are having high MD value and hence, they are efficiently removed prior to the robust estimation. The tipper data computed for the same site (KL33A) is shown in Fig. 16. It shows the real and imaginary parts of T_x and T_y data. The tipper data is averaged using the events selected by MD based selection procedure in this package. We can see smooth curves except in few frequencies for the tipper data.

In Fig. 14, a slight scattering of data points can be observed in frequencies near 0.1 Hz and a drop in coherency values are also seen. Hence, the site is re-processed using the remote reference method implemented in the SigMT. A significant improvement is visible in the output (Fig. 17a) after the application of remote reference method. The scattering near the frequencies 0.1 Hz become smooth after the implementation of the remote reference method. The main change is observable in the predicted coherency values (Fig. 17b). All frequencies in the site shows very good coherencies (near 1.0) for the re-processed data. It shows effectiveness of the remote reference method in the SigMT.

Fig. 18 represents the efficiency of coherency threshold based selection in SigMT. The data with sampling frequency 2 Hz for the site LU16 is processed using the Mahalanobis based selection alone and the robust estimation is shown in Fig. 18a. We can observe scattering and large error bar in both XY and YX components near 0.1 Hz. It is mainly due to the dead band and effect of the incoherent noises in the data. That is the reason for the low predicted coherency (Fig. 18b) for the site. The coherency is less than 0.2 in some frequencies near 0.1 Hz. For this site, we analysed coherency values of all time windows for all target frequencies. For example, argand diagrams of impedance values (xy component) for the target frequencies 0.09 Hz and 0.03 Hz with the predicted coherencies are shown in Fig. 19. For the target frequency 0.09 Hz, it is evident from the figure that very low coherent (<0.3) data points are present in the data, and it forms a small cluster away from the high coherent data points. But there are data points having high coherency above 0.9. Similarly, in 0.03 Hz, most of the data points demonstrates high coherent values (>0.8) in most of the time windows. So, the data is re-processed after selecting the time windows having coherencies above 0.7 for frequencies like in 0.09 Hz and 0.9 for target frequencies like 0.03 Hz. The 'CohThre' was given as:

$$\text{CohThre} = [0.7,0.7,0.7,0.7,0.7,0.7,0.9,0.9,0.9,0.9,0.9,0.9]$$

We can observe significant improvements in the result after reprocessing of data (Fig. 18c). The scattering in the data is reduced and error bars became smaller. The improvement in the predicted coherency (Fig. 18d) is evident for the re-processed data. All frequencies show coherency above 0.5 and high coherency (near 1.0) is seen in low frequencies (<0.1 Hz).

The site LU33A is an example for polarization direction based selection technique. Fig. 20a shows the output of the data sampled with 32 Hz sampling frequency for site LU33A. It is processed with MD based selection alone and the robust estimation. The apparent resistivity and phase data look scattered and having very large error bars. It is observable that XY component has very low coherency values (Fig. 20b) for all the frequencies. The same dataset is re-processed by applying the polarization direction based selection technique. As an example, the polarization direction of magnetic and electric field channels is plotted

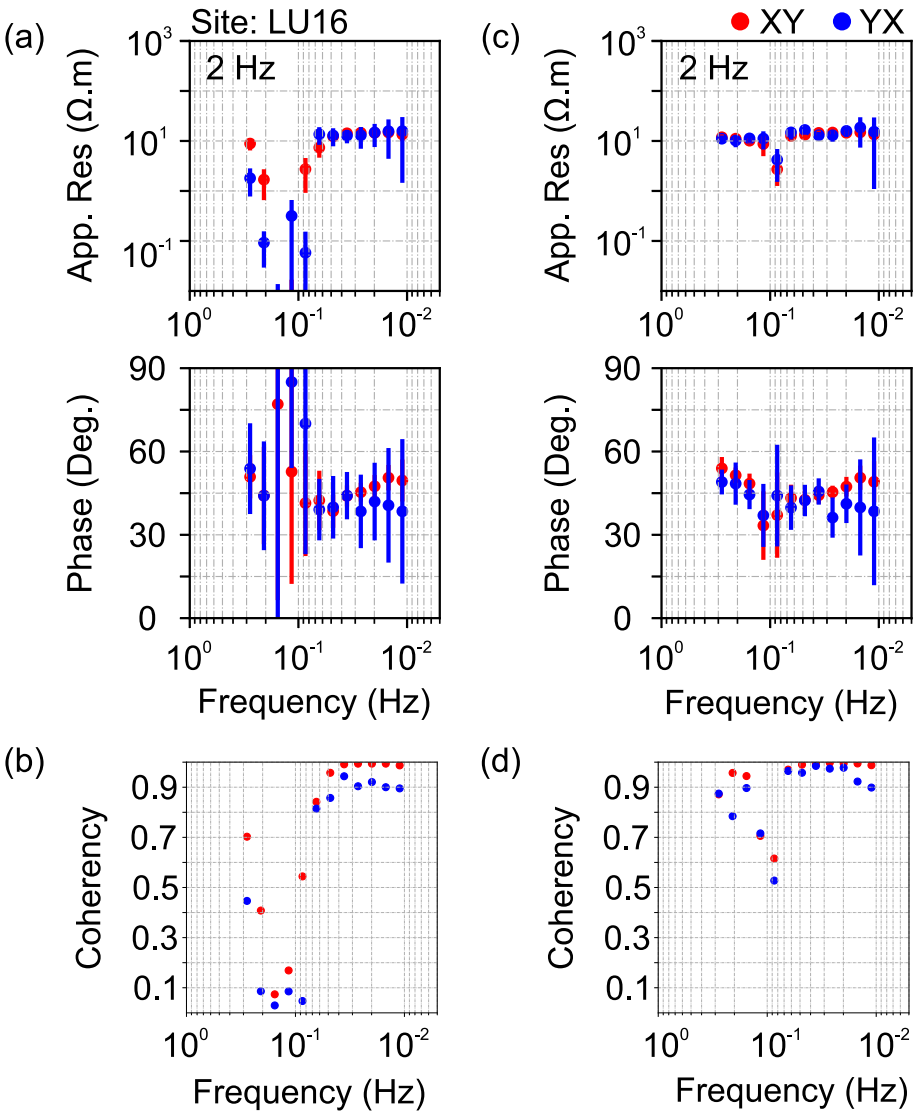


Fig. 18. (a) Apparent resistivity and phase for the site LU16 using Mahalanobis based data selection only. (b) Predicted coherency for the data shown in Fig. 18a. (c) Apparent resistivity and phase for the site LU16 after coherency threshold based data selection and Mahalanobis based data selection. (d) Predicted coherency for the data shown in Fig. 18c.

for the target frequency 2.17 Hz is shown in Fig. 21. The electric field channel is strongly polarized in a range of (−10,10) from near the 200th time window to the end of the data. Hence, we removed all the time windows having electrical polarization direction between [−10,10] and re-processed the data. The following code is used for this purpose:

```
pdlim = [-10,10]
```

The apparent resistivity and phase of the re-processed data is shown in Fig. 20c. It is observable that the apparent resistivity and phase curves have become smoother compared to with Fig. 20a. The large error bars became significantly reduced in the re-processed data. Coherency has greatly improved after the polarization direction based selection (Fig. 20d). The data up to 1 Hz shows improved predicted coherency in comparison with the prior data (Fig. 20b). The data points show coherency above 0.8 till the 1 Hz. Then the coherency tends to decrease, probably because of incoherent noises in the data. Because of the low coherency, larger error bars can be seen in the frequencies less than 1 Hz. This example demonstrates the efficiency of polarization based data selection in the SigMT.

6. Conclusion

The MT data acquisition has become a challenge in the modern day due to the unavailability of noise-free locations. In many cases, the MT site selection is compromised, and data are collected from the noisy environments. Hence, the advancement in the MT data processing approaches and understanding of noises have become a necessity. The MT data processing is often a time-consuming job due to the requirement of the manual inspection of the time series. In this study, we have compiled the existing data processing tools and built an open-source Python package, 'SigMT' for the automated MT data processing. This Python package enables the MT data processing without any manual time series inspection. For this purpose, we have used Mahalanobis distance based data selection method. The Mahalanobis based data selection is used for all the datasets to reject most of the outliers in the dataset. In addition, different data selection tools such as coherency based data selection, polarization direction based data selection are implemented in the SigMT. The application of these data selection tools has been explained with examples in this paper and it shows the efficacy of SigMT for the MT data processing.

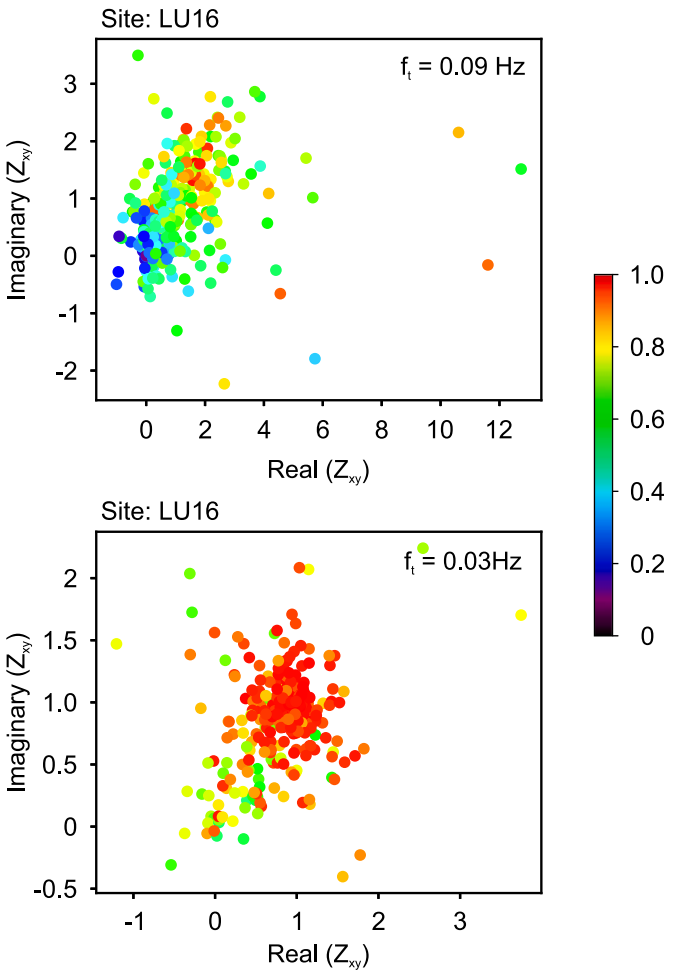


Fig. 19. Predicted coherency for all time windows at the target frequencies 0.12 and 0.02 Hz for the site LU16.

7. Future scope

This Python package enables the MT data processing without any manual time series editing. It incorporates different data selection tools such as Mahalanobis, coherency threshold, and polarization direction based data selection tools. It is sufficient for most of the cases except when the majority of time series is severely contaminated by noise. This processing package will be the basic platform for our future research to understand electromagnetic noises and advancement in MT data processing. It is built in an easy-to-use manner, hence anyone can run the code without much programming knowledge and can perform experiments in it. In the present form, SigMT supports only '.ats' file format. But it will be extended to other data types in the future updates.

Authorship contribution statement

K.S. Ajithabh: Development of the Python package, manuscript preparation, data acquisition in Himalaya and processing. Prasanta K. Patro: Conceptualization and supervision of the developmental work, supervision of data acquisition in Himalaya and processing, manuscript preparation.

9. Code availability

SigMT is available at: https://github.com/ajithabhks/SigMT

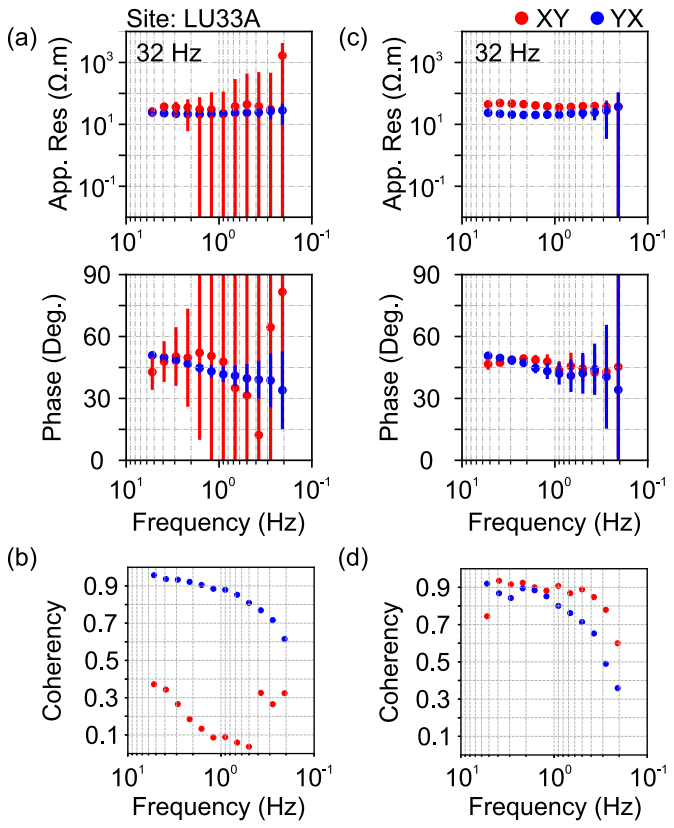


Fig. 20. (a) Apparent resistivity and phase for the site LU33A using Mahalanobis based data selection only. (b) Predicted coherency for the data shown in Fig. 20a. (c) Apparent resistivity and phase for the site LU33A after polarization direction based data selection and Mahalanobis based data selection. (d) Predicted coherency for the data shown in Fig. 20c.

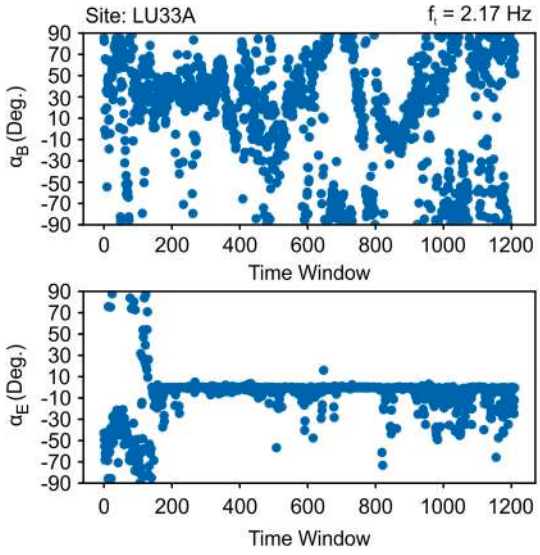


Fig. 21. Polarization direction for magnetic field (α_B) and electric field (α_E) for all time windows at the target frequency 2.17 Hz for the site LU33A.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

We thank Dr. V.M. Tiwari, Director, CSIR – NGRI for giving permission to publish the work. This study forms part of the PhD work of KSA. The data used in this study was acquired as part of the project 'Synthesis of Earthquake Hazard Scenario in NW Himalaya by Investigating the Multi-Scale Variations in Structural and Seismotectonic Assemblages (SHIVA)'. We thank Dr. Manoj C. Nair and Mr. Mahesh Narayanan for the fruitful discussions regarding the computational aspects of MT data processing. We acknowledge Dr. Bernhard Friedrichs for sharing details about MFS06e induction coil magnetometer and answering some questions. We thank both the editor and the reviewer for their suggestions which has improved the clarity in the manuscript. This research is supported through the projects MLP-0001-28-FBR-1, MLP-6402-28 (BPK) and MLP-FBR-003.

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